Relational-WorDepth: Object-Level Depth Ordering for Monocular Depth Estimation

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Abstract

Monocular depth estimation is inherently ambiguous due to the loss of 3D information in 2D images. WorDepth mitigates this ambiguity by incorporating text captions as a language prior to recover metric-scale depth. However, language supervision is used only as a global cue and does not explicitly enforce object-level spatial relationships.We propose Relational-WorDepth, which introduces explicit object-level structural supervision. By constructing pairwise depth relations between object instances and enforcing relative depth ordering with a ranking-based loss, our method directly constrains scene geometry during training. Experiments on NYU-Depth-v2 demonstrate consistent improvements over the baseline and improved geometric consistency in predicted depth maps.

1. Introduction

Estimating 3D scene geometry from a single image is inherently ambiguous, as infinitely many 3D scenes can project to the same 2D observation. WorDepth[4] addresses this ambiguity by incorporating text captions as a language prior within a variational framework, enabling metric-scale depth estimation by leveraging semantic cues about object sizes and typical scene layouts.

However, WorDepth exhibits structural limitations in how language supervision is utilized. Textual information is treated primarily as a global contextual prior, without explicitly enforcing object-level spatial relationships such as relative depth ordering. Moreover, the variational bottleneck injects language information only as a weak preference, rather than as an explicit geometric constraint, which can lead to inconsistencies in the predicted relative depth between scene elements.

To address these limitations, we propose Relational-WorDepth, which augments implicit language priors with explicit structural supervision. Our approach models object-level depth relationships using object instances detected by YOLO-World[1] and precise pixel-level masks generated by SAM2[2]. We further introduce a Relational Depth Loss that directly enforces relative depth ordering during training. By combining semantic priors with explicit geometric constraints, Relational-WorDepth improves the structural consistency of monocular depth estimation.

2. Methodology

Following WorDepth[4], we address supervised monocular depth estimation by learning a mapping h that predicts a dense depth map from an RGB image and a text caption. While WorDepth uses language as a variational prior to resolve scale ambiguity, it does not explicitly enforce object-level spatial relationships. We propose *Relational-WorDepth*, which introduces explicit object-level depth relations and a relational loss to improve geometric consistency.

2.1. Constructing Structural Constraints

To supervise relative depth ordering, object masks and relational labels are generated offline.

Object Instance Segmentation. We detect object bounding boxes for the 40 NYU-Depth-v2[3] classes using YOLO-World[1] and use them as prompts for SAM2[2] to obtain binary object masks $M = \{M_1, \ldots, M_N\}$.

Generation of Structural Relationships. Scene structure is represented as a set of pairwise depth relations $\mathcal{R}$. Relations are assigned based on median depth differences exceeding a threshold τ (e.g., 0.3m) and restricted to spatially neighboring object pairs. Each relation is categorized as `front` or `behind`.

2.2. Relational Depth Loss

To enforce the constraints in $\mathcal{R}$, we introduce a ranking-based relational loss. For each object i, we compute a representative depth $\bar{y}_i$ as the mean predicted depth within its mask M_i:

$$\bar{y}_i = \frac{1}{|M_i|} \sum_{p \in M_i} y(p), \tag{1}$$

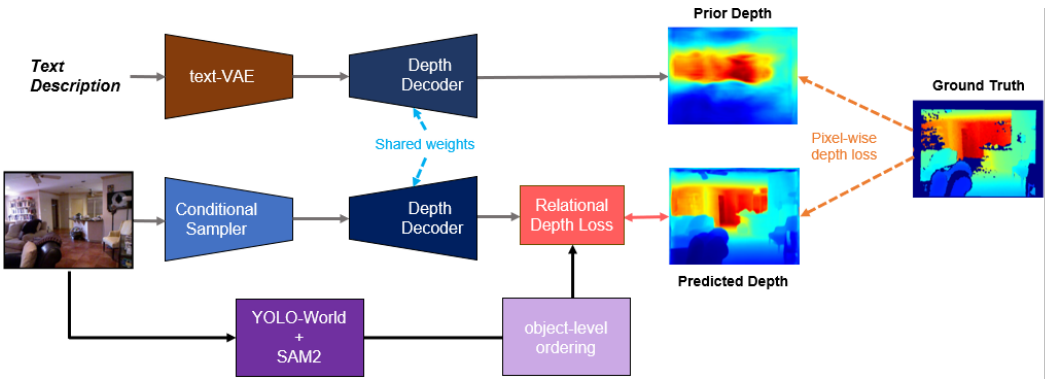

Figure 1. **The proposed WorDepth architecture with relational supervision.** The additional branch (orange) enforces object-level depth ordering.

where p denotes pixel coordinates.

For each relation $(i, j, r) \in \mathcal{R}$ indicating that object i is in front of object j, we enforce $\bar{y}_i < \bar{y}_j$ using a margin-based penalty:

$$\mathcal{L}_{rel} = \frac{1}{|\mathcal{R}|} \sum_{(i,j,r)\in\mathcal{R}} \mathrm{ReLU}(\bar{y}_i - \bar{y}_j + m_r). \qquad (2)$$

Here, m_r denotes a margin that enforces a minimum separation between the predicted depths of two related objects.

3. Experiments

We evaluate the proposed Relational-WorDepth on the NYU-Depth-v2[3] benchmark. The goal of our experiments is to analyze how explicitly enforcing object-level relative depth relationships affects metric depth estimation performance and geometric consistency.

Dataset. We conduct experiments on the NYU-Depth-v2 dataset, which contains 464 indoor scenes with aligned RGB and depth images. Following standard practice, we use the official train/test split and report results on the 654-image test set.

Implementation Details. All models are trained under identical settings, with Relational-WorDepth differing only by the proposed relational loss. We report Abs Rel, RMSE, and δ_1 accuracy for evaluation.

3.1. Quantitative Results

Table 1 compares WorDepth and Relational-WorDepth on the NYU-Depth-v2 test set. Results are reported at the same training stage for a fair comparison. Relational-WorDepth achieves consistent improvements across all metrics.

3.2. Qualitative Comparisons

Figure 2 compares qualitative results of WorDepth and Relational-WorDepth. While the baseline often exhibits blurred depth transitions and structural inconsistencies near object boundaries, Relational-WorDepth produces sharper discontinuities and more coherent object-level structures.

Method	Abs Rel ↓	RMSE ↓	$\delta < 1.25$ ↑
WorDepth	0.137	0.478	0.833
Rel.-WorDepth (Ours)	**0.131**	**0.470**	**0.840**

Table 1. **Quantitative results on the NYU-Depth-v2 benchmark.** WorDepth serves as the baseline model, and Relational-WorDepth extends it by incorporating explicit object-level relational supervision.

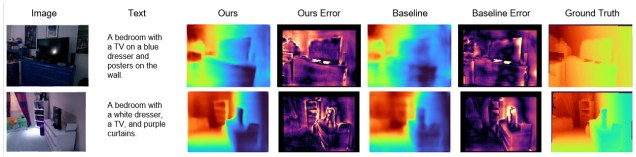

Figure 2. **Qualitative comparison of depth estimation.** From left to right: Input image, Text prompt, Ours, Error map (Ours), Baseline, Error map (Baseline), and Ground Truth.

4. Conclusion

We presented Relational-WorDepth, a monocular depth estimation framework that augments language-based priors with explicit object-level relational supervision. By enforcing relative depth ordering through a relational loss, our method improves geometric consistency while preserving metric depth accuracy.

Although the quantitative improvements over the baseline are moderate, they are consistent across all evaluation metrics. This indicates that relational constraints effectively complement variational language priors by correcting structural inconsistencies that are not explicitly constrained by language alone.

References

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